POSTER

Using The HeartRateLab.jl for Normative Evaluation of Heart Rate Variability from Open Datasets

Alberto Barradas (abcsds) — Heart Rate Variability analysis, entirely in Julia

github.com/abcsds/HeartRateLab.jl

Abstract

A common method in the evaluation of human biosignals is comparing experimental signals against datasets of normative values. Thanks to the open scientific resources for heart rate analysis, and the resilient network of scientific programming functionality that the Julia community enables, the normative analysis of Heart Rate Variability (HRV) can be done **completely in Julia**. By contrasting features in windows of **360 heart beats** against normative datasets we identified that meditators from the *Heart Rate Oscillations during Meditation* dataset (PhysioNet) presented indicators of increased **Mean IBI** ($z = +1.02$) and **pNN50** ($z = +1.32$), with smaller but consistent increases in other features common in resonant-frequency training that remained under 1σ: **RMSSD** ($z = +0.94$), **SDNN** ($z = +0.90$). Results demonstrate how correctly implemented open science can answer contextual questions such as: *"Is this new experimental data normal?"*

Why context matters

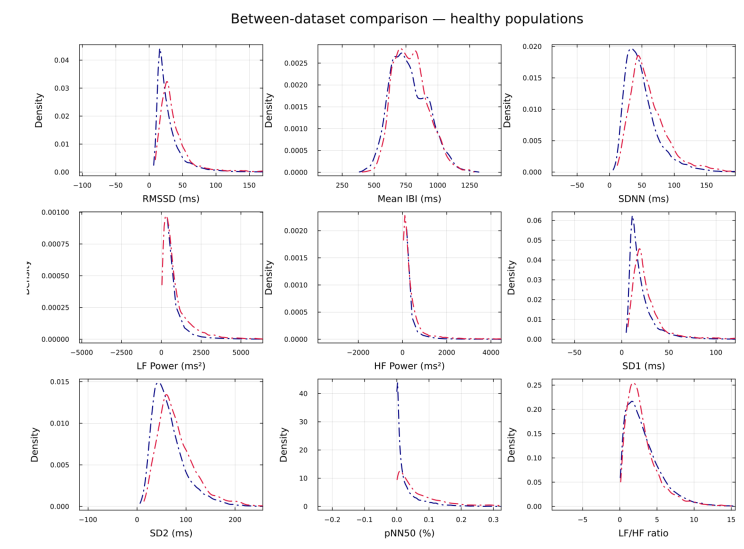
We contrast a time series of **Inter-Beat Intervals (IBIs)** against the feature distributions of existing open datasets, emphasising the **contextual nature** of scientific inquiry. This capability is a novel addition implemented specifically for HeartRateLab.jl.

An example contrasting meditation recordings demonstrates current capabilities and illustrates functionality that aids **hypothesis testing**, **feature engineering**, and a deeper understanding of recordings — including use as a **personal daily HRV monitor** that norms your own signals against a healthy population.

Method — a normative pipeline, end to end in Julia

1. Read IBIs (`read_txt` / `read_wfdb` / `read_xdf`).
2. Preprocess: replace bio-outliers, ectopic & zero beats; interpolate `NaN`s.
3. Extract **53 HRV features** (time / frequency / nonlinear) via the `@register` + `@memoize` registry.
4. **Window** each recording: 360 beats, 120-beat stride.
5. Fit a **normative prior** per feature (Normal / Gamma / Beta / LogNormal) by MLE over **56,472 windows from 66 healthy subjects** (PhysioNet `nsrdb` + `nsr2db`).
6. Score any new window: percentile $= F_{\text{prior}}(x)$, $\quad z_{\text{equiv}} = \Phi^{-1}(\text{percentile})$.

*68 / 95 / 99.7% dispersion bands below come from the fitted priors — these are **window-level** (windows within a recording are correlated), so a z is descriptive, not a per-subject inferential test.*

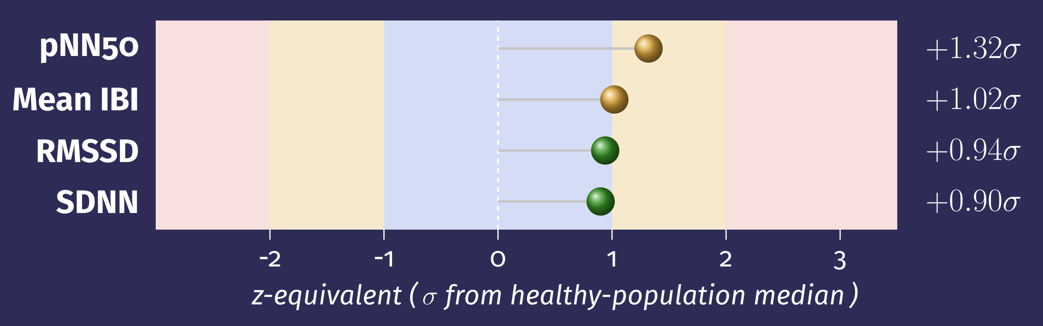


Validity check: the two healthy reference databases (`nsrdb`, `nsr2db`) agree feature-by-feature, so the pooled prior is well-defined.

Results — two contextual cases

Case A — Meditation cohort

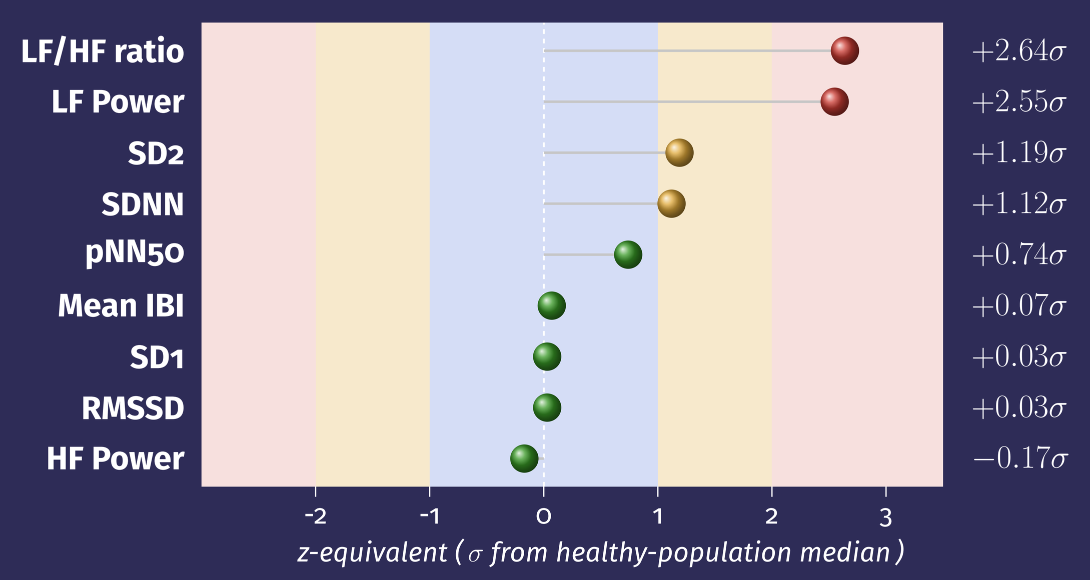
PhysioNet *Heart Rate Oscillations during Meditation* · 58 records · median window vs healthy prior.



Meditators trend to **longer IBIs** and **elevated pNN50**; short-term vagal tone (RMSSD, SDNN) is **modestly elevated but under 1σ** — a mild, coherent shift, not pathology.

Case B — Anonymous longitudinal participant

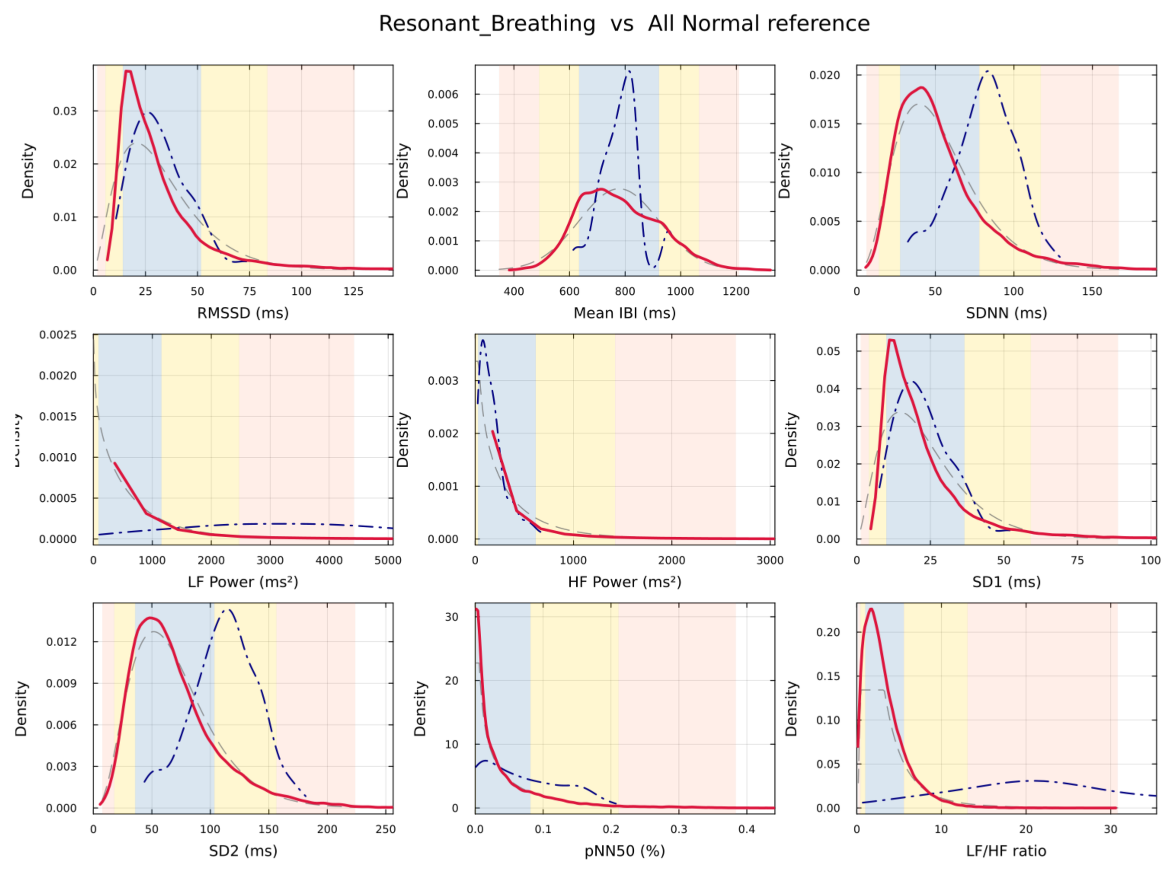
A **daily HRV monitor**: 148 windows across **88 recording dates**, 2019–2025, vs the same healthy prior.



The resonance signature: the $\sim$0.1 Hz **LF band** is strongly elevated (LF power & the LF/HF ratio derived from it, $> +2.5\sigma$) while parasympathetic & short-term metrics (RMSSD, HF, SD1, Mean IBI) stay near the healthy median. The monitor's z-scores *track the practice* — consistent with slow, resonant breathing.

Single participant; 148 correlated windows (descriptive, not inferential). Features are not independent — SD1 $\equiv$ RMSSD$/\sqrt{2}$, SD2 tracks SDNN, LF/HF derives from LF power — so nine labels span $\sim$5 axes.

Participant vs. healthy population — all nine features



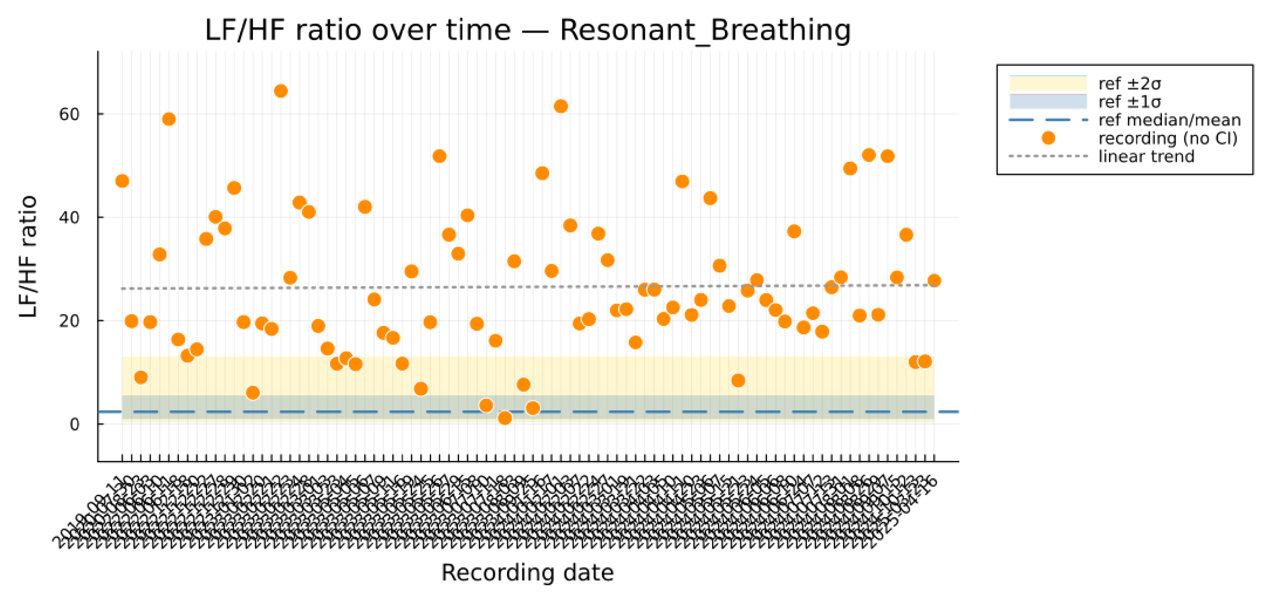
How to read. Each panel is one HRV feature. **Blue dash-dot** = this participant's 148 windows; **red** = the fitted *parametric* prior; grey dashed = empirical KDE of the same pooled data (fit-quality check). Shaded bands = central **68 / 95 / 99.7%**.

Right of reference (elevated): LF Power, LF/HF, SD2, SDNN.
Inside reference (typical): RMSSD, HF, SD1, Mean IBI, pNN50.

The σ-equivalent is a percentile re-expressed on a standard normal (e.g. LF Power $+2.55\sigma \equiv$ 99.5th percentile of healthy windows), not standard deviations of the skewed feature.

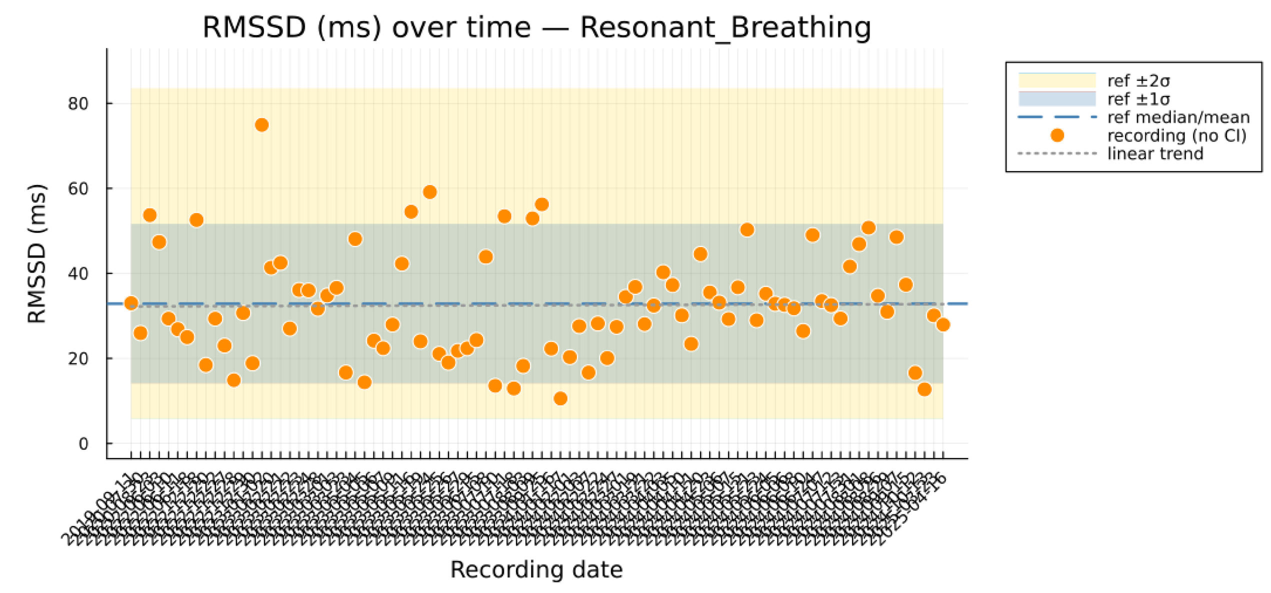
→ A selective boost in the low-frequency band — consistent with slow, resonant breathing, not a change in short-term vagal markers.

LF/HF over time — above $+2\sigma$



Nearly every session rides *above* the normative $\pm 2\sigma$ band — the practice is visible for years.

RMSSD over time — inside $\pm 1\sigma$



The same recordings' short-term vagal tone stays *within* the healthy $\pm 1\sigma$ band — typical, as expected.

"Is this new experimental data normal?" HeartRateLab.jl answers it **in context** — fitting open-data priors and scoring any recording, from a clinical cohort to your own daily practice, with one reproducible pipeline.

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abcsds/HeartRateLab.jl
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